

Class #3

É. Savin

Numerical
data

Cavity filled
with water

Cavity filled
with air

Acoustic cavity coupled to a plate

MG3416–Advanced Structural Acoustics - Class #3

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Outline

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1 Numerical data

2 Cavity filled with water

3 Cavity filled with air

Numerical data

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Plate:

$$a = 17 \text{ cm}, b = 15 \text{ cm}$$

$$E_1 = 1.62 \cdot 10^{11} \text{ Pa}$$

$$\nu_1 = 0.3$$

$$\varrho_1 = 31.2 \text{ kg/m}^3$$

$$\eta_{d1} = 0.8$$

$$F = 1 \text{ N}$$

$$x_0 = 5.1 \text{ cm}, y_0 = 4.5 \text{ cm}$$

$$f_c = 64915 \text{ Hz}$$

Cavity with water:

$$L = 31 \text{ cm}$$

$$c_2 = 1500 \text{ m/s}$$

$$\varrho_2 = 1000 \text{ kg/m}^3$$

$$\eta_{d0} = 0.001 \text{ kg/m.s}$$

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Plate:

$$a = 17 \text{ cm}, b = 15 \text{ cm}$$

$$E_1 = 1.7 \cdot 10^{11} \text{ Pa}$$

$$\nu_1 = 0.3$$

$$\varrho_1 = 31.2 \text{ kg/m}^3$$

$$\eta_{d1} = 0.8$$

$$F = 1 \text{ N}$$

$$x_0 = 5.1 \text{ cm}, y_0 = 4.5 \text{ cm}$$

$$f_c = 3256 \text{ Hz}$$

Cavity with air:

$$L = 31 \text{ cm}$$

$$c_2 = 340 \text{ m/s}$$

$$\varrho_2 = 1.3 \text{ kg/m}^3$$

$$\eta_{d0} = 1.705 \cdot 10^{-3} \text{ kg/m.s}$$

Cavity filled with water

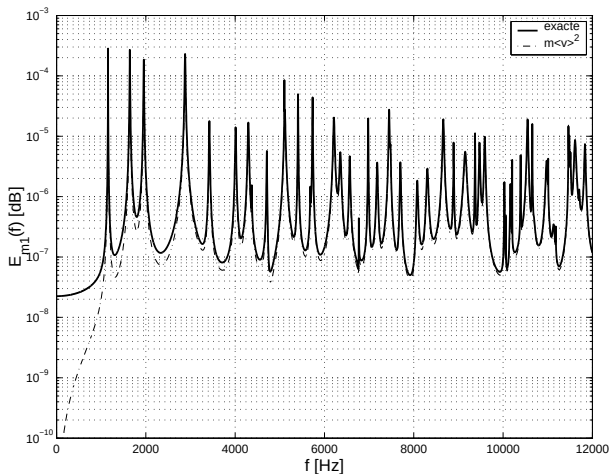
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Mechanical energy of the plate
 $\text{dB}_{\text{réf}} = 10 \times \log_{10}(1 \text{ kg.m}^2/\text{s}^2)$

Cavity filled with water

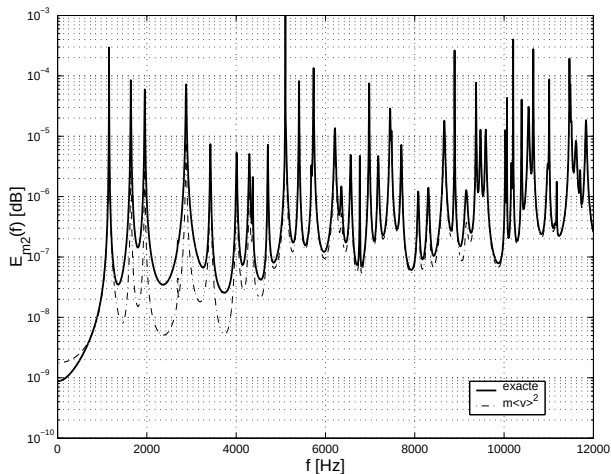
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Acoustical energy of the cavity
 $\text{dB}_{\text{réf}} = 10 \times \log_{10}(1 \text{ kg.m}^2/\text{s}^2)$

Cavity filled with water

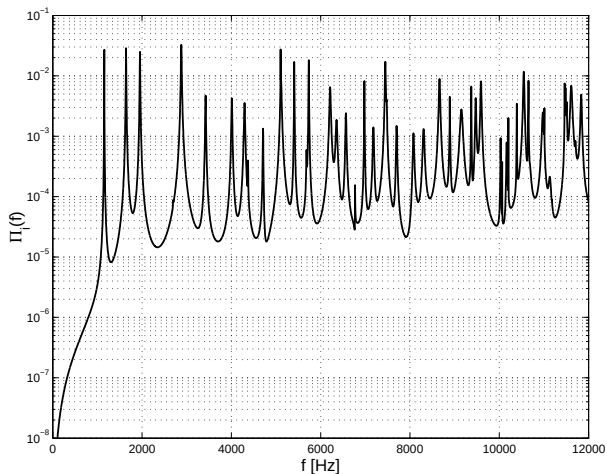
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Total input power in the coupled plate/cavity system

Cavity filled with water

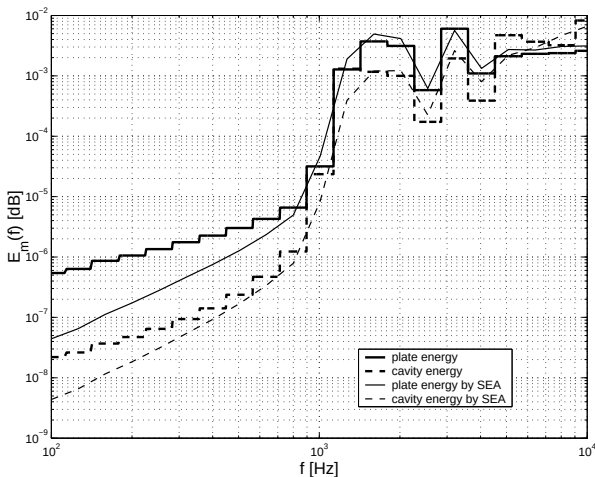
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Comparisons of the total energies computed analytically and by an SEA approach

Cavity filled with air

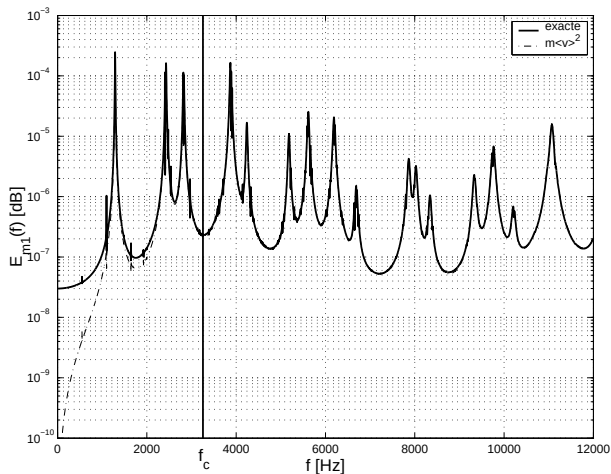
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Mechanical energy of the plate
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Cavity filled with air

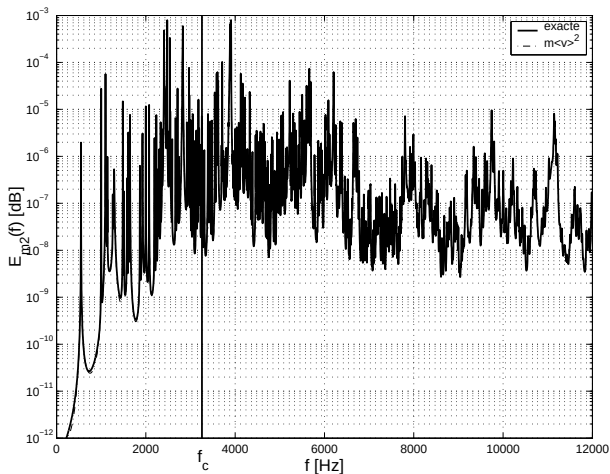
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Cavity filled with air

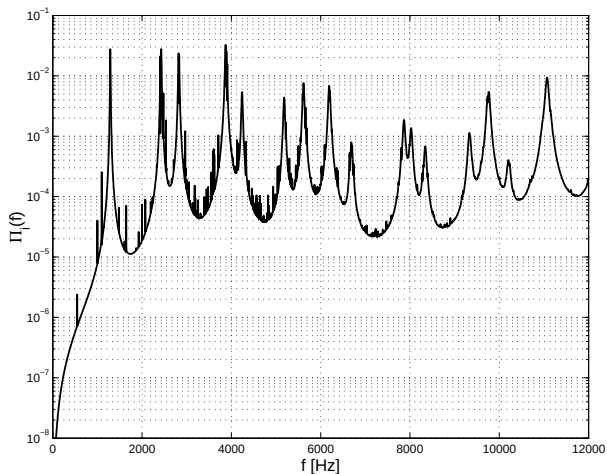
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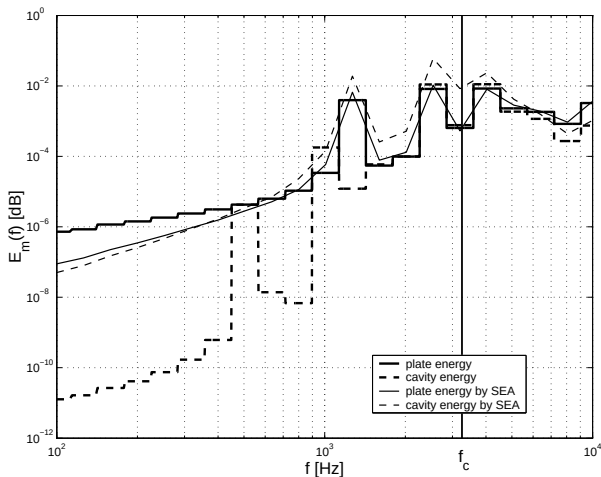
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